

GOODRIDGE 12 NNW, MN, USA

WMO: 726470

Lat: 48.3055N

Lon: 95.8744W

Elev: 351

StdP: 97.18

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 04994

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.5	(c) -29.0	(d) -34.7	(e) 0.1	(f) -31.4	(g) -32.2	(h) 0.2	(i) -28.8	(j) 7.1	(k) -13.8	(l) 6.2	(m) -11.8	(n) 0.9	(o) N/A	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	29.3	20.9	27.8	19.7	26.4	18.9	22.6	27.6	21.3	26.2	20.2	24.9	2.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	16.0	25.9	19.5	14.8	24.7	18.4	13.8	23.3	68.0	27.8	63.4	26.3	59.3	24.8	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.7	(b) 4.9	(c) 4.4	(e) -36.4	(f) 32.3	(g) 3.5	(h) 1.7	(i) -39.0	(j) 33.6	(k) -41.0	(l) 34.6	(m) -43.0	(n) 35.6	(o) -45.5	(p) 36.9		
(a) 5.7	(b) 4.9	(c) 4.4	(e) -35.5	(f) 24.7	(g) 3.8	(h) 1.1	(i) -38.2	(j) 25.4	(k) -40.5	(l) 26.1	(m) -42.6	(n) 26.7	(o) -45.4	(p) 27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-14.7	-13.4	-4.4	4.3	11.2	16.5	18.9	17.3	13.2	5.9	-3.0	-11.2	(6)
(7)		DBStd	13.27	8.65	7.50	7.57	5.73	4.97	3.40	3.10	3.47	4.74	5.04	6.52	7.28	(7)
(8)		HDD10.0	3320	767	657	449	184	45	1	0	1	21	148	390	658	(8)
(9)		HDD18.3	5539	1025	890	706	422	226	72	30	63	164	385	639	917	(9)
(10)		CDD10.0	936	0	0	2	12	83	197	277	226	116	22	1	0	(10)
(11)		CDD18.3	113	0	0	0	0	6	18	49	29	10	1	0	0	(11)
(12)		CDH23.3	1273	0	0	0	3	86	211	519	321	118	15	0	0	(12)
(13)		CDH26.7	259	0	0	0	0	19	32	116	67	23	2	0	0	(13)
(14)	Wind	WSAvg	1.9	2.0	2.0	2.1	2.2	2.1	1.8	1.5	1.4	1.7	1.9	2.0	1.9	(14)
(15)	Precipitation	PrecAvg	599	16	13	21	36	78	99	97	89	64	49	26	19	(15)
(16)		PrecMax	722	28	24	67	65	183	225	154	144	137	129	83	35	(16)
(17)		PrecMin	421	4	2	5	16	23	22	22	22	12	7	2	7	(17)
(18)		PrecStd	93	8	7	15	16	42	50	36	34	35	30	22	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	5.2	18.1	22.7	29.1	29.6	31.3	30.8	29.3	25.6	14.9	4.3	(19)
(20)			MCWB	1.7	2.8	12.3	10.8	17.6	20.4	22.6	22.0	19.9	15.2	9.1	2.7	(20)
(21)		2%	DB	1.2	1.9	12.3	19.5	25.8	27.4	29.5	28.6	26.7	20.7	10.9	1.9	(21)
(22)			MCWB	0.1	0.3	8.0	10.5	15.4	18.6	22.1	20.3	18.1	12.9	7.0	0.2	(22)
(23)		5%	DB	-0.4	0.2	8.0	16.8	23.1	25.8	27.8	26.7	24.3	17.1	8.2	0.3	(23)
(24)			MCWB	-1.5	-1.1	4.7	9.0	13.8	17.7	20.9	19.7	17.1	11.4	5.4	-0.8	(24)
(25)		10%	DB	-2.4	-2.1	5.0	14.3	20.6	24.1	26.3	25.0	21.7	14.3	5.6	-1.4	(25)
(26)			MCWB	-3.4	-3.2	2.4	7.7	12.7	17.0	19.8	18.6	16.0	9.5	3.4	-2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	3.5	14.8	12.3	19.1	22.2	24.7	23.4	22.0	15.7	10.0	3.5	(27)
(28)			MCDWB	3.9	6.0	17.9	17.9	26.2	26.3	29.0	29.5	26.2	24.3	14.1	4.3	(28)
(29)		2%	WB	0.6	0.7	9.8	10.2	17.5	20.2	23.0	22.2	19.8	14.0	7.6	1.0	(29)
(30)			MCDWB	1.5	2.1	13.1	17.1	23.8	25.4	27.8	27.5	23.9	19.5	10.7	2.2	(30)
(31)		5%	WB	-1.2	-0.8	5.6	8.9	15.7	18.9	21.6	20.8	18.1	12.0	5.3	-0.4	(31)
(32)			MCDWB	0.0	0.4	9.1	15.3	20.9	24.5	26.5	25.5	22.6	16.2	7.8	0.8	(32)
(33)		10%	WB	-3.3	-2.9	3.0	7.4	14.3	17.9	20.5	19.8	16.4	10.1	3.3	-2.1	(33)
(34)			MCDWB	-2.4	-1.8	5.9	12.9	19.1	23.2	25.3	24.1	20.8	14.4	5.3	-1.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	11.0	10.6	11.8	12.4	11.6	12.5	13.1	12.5	10.2	8.8	8.6	(35)
(36)			MCDBR	9.1	10.0	13.1	16.5	16.4	14.4	13.9	14.9	15.4	15.0	11.9	9.1	(36)
(37)			MCWBR	8.7	8.5	9.1	9.4	8.3	7.8	7.5	8.0	8.6	8.2	7.9	7.2	(37)
(38)		5% WB	MCDWB	9.7	9.5	13.3	15.5	13.0	13.0	12.2	13.1	13.0	12.6	10.9	8.0	(38)
(39)			MCWBR	8.5	8.0	9.3	9.3	7.4	7.6	7.4	7.5	8.0	7.4	7.5	6.9	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.271	0.306	0.294	0.346	0.362	0.376	0.379	0.377	0.348	0.303	0.276	0.263	(41)	
(42)		taud	2.281	2.175	2.415	2.366	2.381	2.367	2.372	2.383	2.459	2.553	2.507	2.378	(42)	
(43)		Ebn at Noon	798	838	930	905	899	885	877	862	858	854	807	770	(43)	
		Edh at Noon	73	103	96	112	115	118	116	110	92	71	59	59	(44)	
(44)	All-Sky Solar Radiation	RadAvg	1.49	2.71	3.88	4.83	5.42	5.78	6.20	5.17	3.81	2.30	1.33	1.05	(44)	
(45)		RadStd	0.16	0.22	0.39	0.43	0.33	0.44	0.28	0.36	0.28	0.34	0.15	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page